

Networks and Systems: MM7

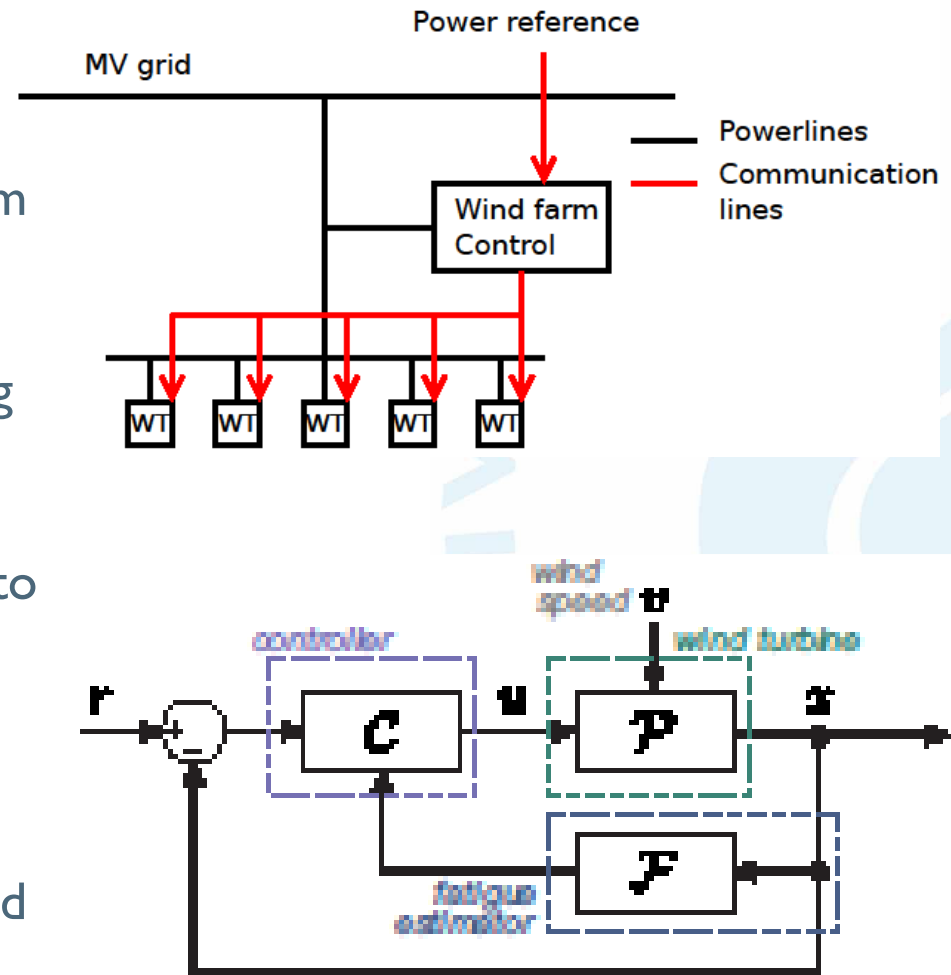
Hans-Peter Schwefel

- **Mm2** Layer-2 Example Technologies: Field-buses CAN and Flexray (hps)
- **Mm3** Clock Synchronisation & impact of clock offsets on sensor data (hps)
- **Mm4 & 5** Network Calculus I & II (HS)
- **Mm6** TCP Modeling and Performance Analysis (hps)
- **Mm7** Performance implications of data access schemes (hps)
- **Mm8** Discrete Event Simulation (HS)
- ...

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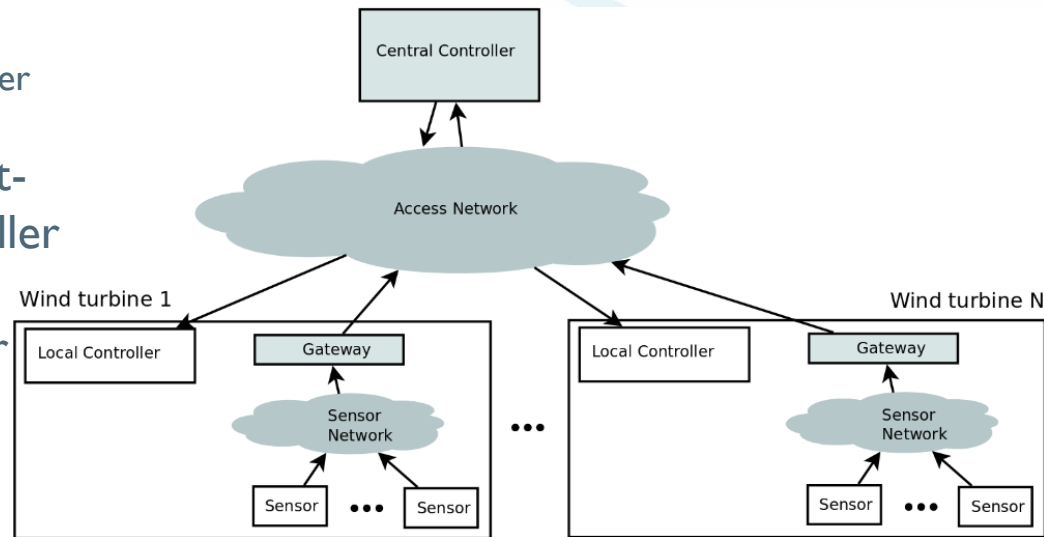
Motivation - Wind-farm scenario

- Control of a wind-farm from a central controller
- Attempt to reduce damage wind turbine sustains during usage while maintaining a power reference
- Uses state of wind turbine to estimate fatigue
- Controller acts periodically with period of 150 ms
- Performance metric is accumulated damage of wind turbine $\rightarrow$ to be minimized

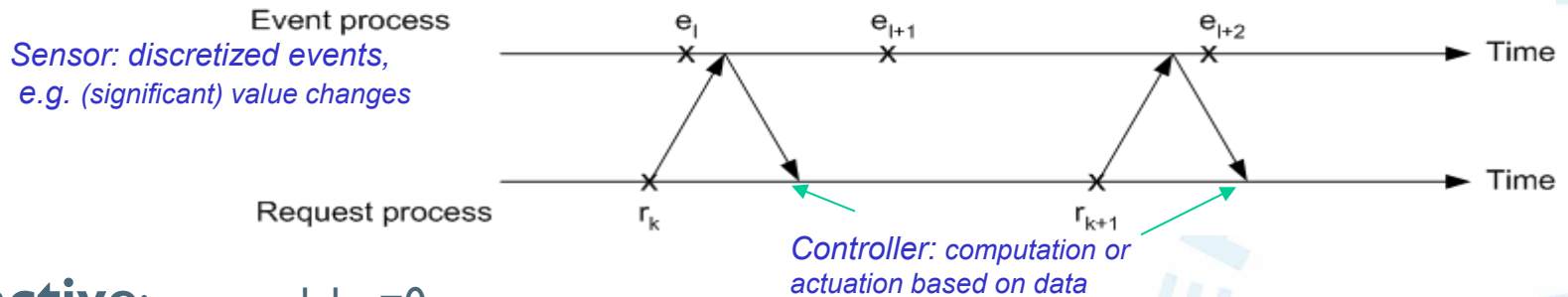


Wind-farm communication network

- Central controller communicates with each wind turbine over Access Network
 - Examples of technologies: fiber optics, LTE, 3G, 2G, PLC
- Local controller acts on set-points from central controller
- Sensors periodically send measurements over Sensor Network to gateway
 - Examples of technologies: Fieldbus, CANbus, WSN
- 3 sensors on each wind turbine
 - Torque x 2, angle rotation
- Gateway forwards sensor information

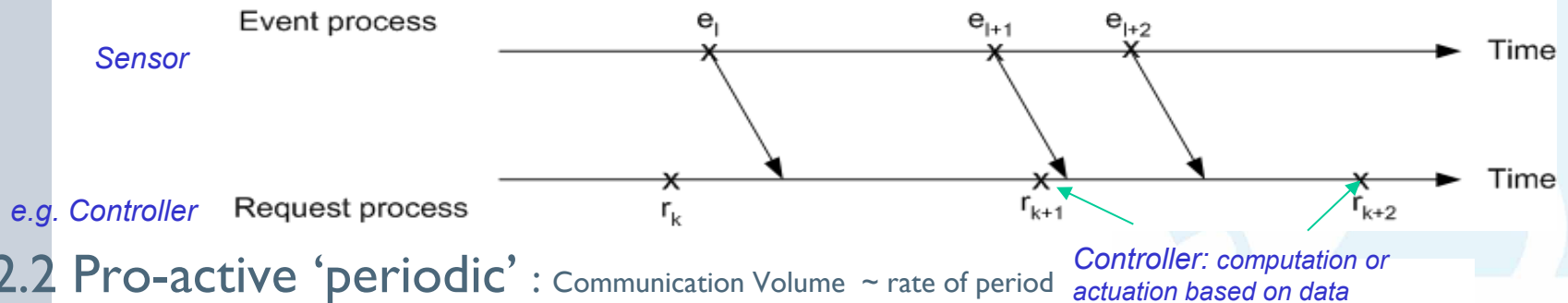


1. Reactive/on-demand: access delay > 0 , Communication Volume $\sim$ request rate

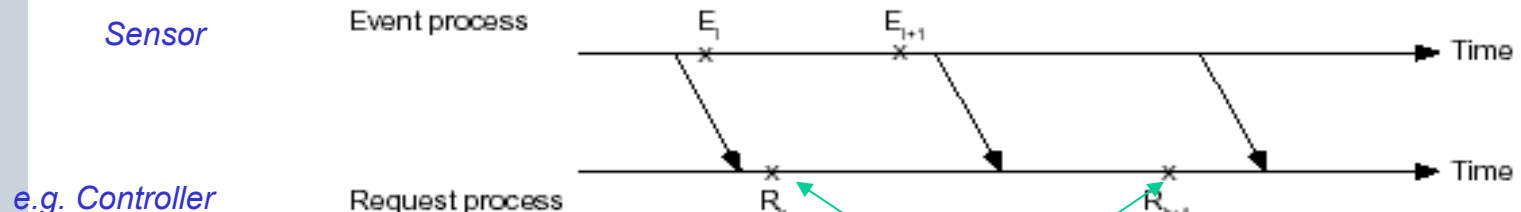


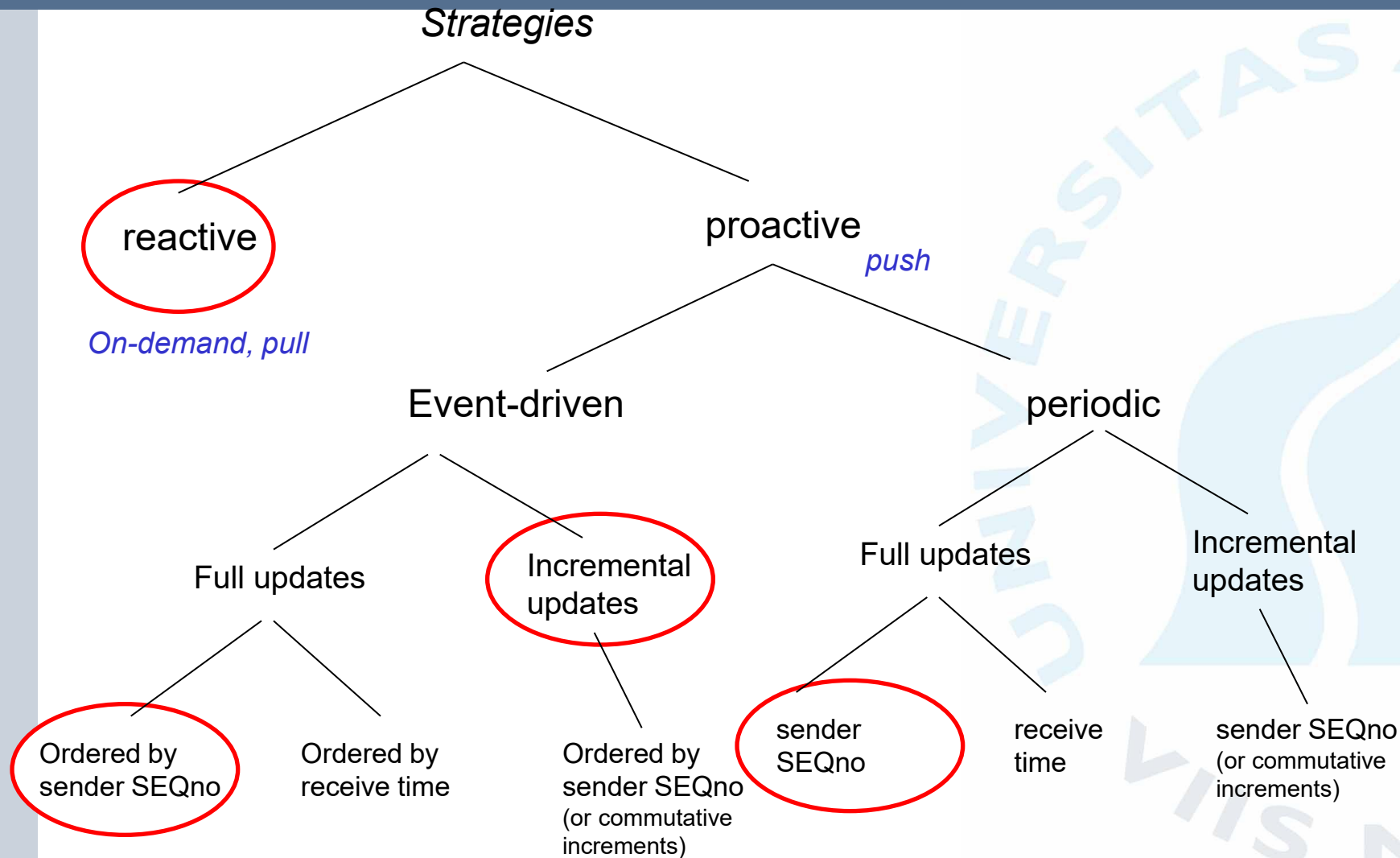
2. Pro-active: access delay=0

2.1 Pro-active event-driven: Communication Volume $\sim$ event-rate



2.2 Pro-active 'periodic' : Communication Volume $\sim$ rate of period





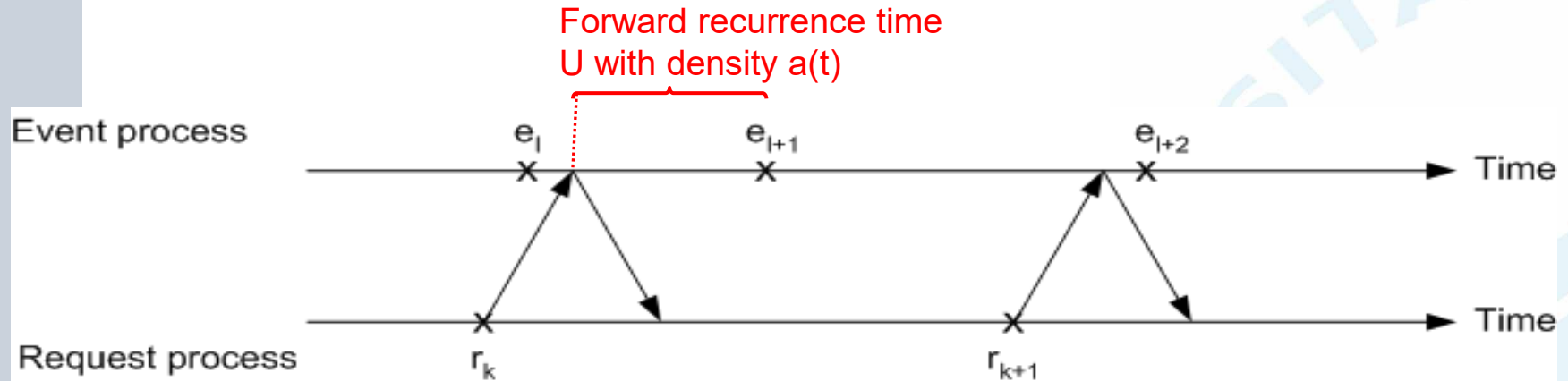
Participating Processes

1. Event Process $\mathcal{E}$ (when Poisson, with rate λ) – significant change of value at sensor
2. Request process $\mathcal{R}$ (with rate μ) - requests of sensor information by controller
3. Downstream Delay $\mathcal{D}$ (when Poisson, with rate ν) - from sensor to controller
4. Upstream Delay $\mathcal{U}$ (only in case of reactive approach) - from controller to sensor
5. Period of pro-active periodic approach $\mathcal{P}$ (when Poisson, with rate τ)

Assumptions

- All processes are mutually independent
→ request process irrelevant for stationary mmPr
- Message re-ordering can occur in network, but requestor can put messages back in order (e.g. using sequence numbers)
- The information element never changes back to an earlier value, e.g. as for monotonous behavior

Interest in mismatch probability = mmPr = probability that value known by request process (controller) does not match the value at the sensor at the time instant when using the information



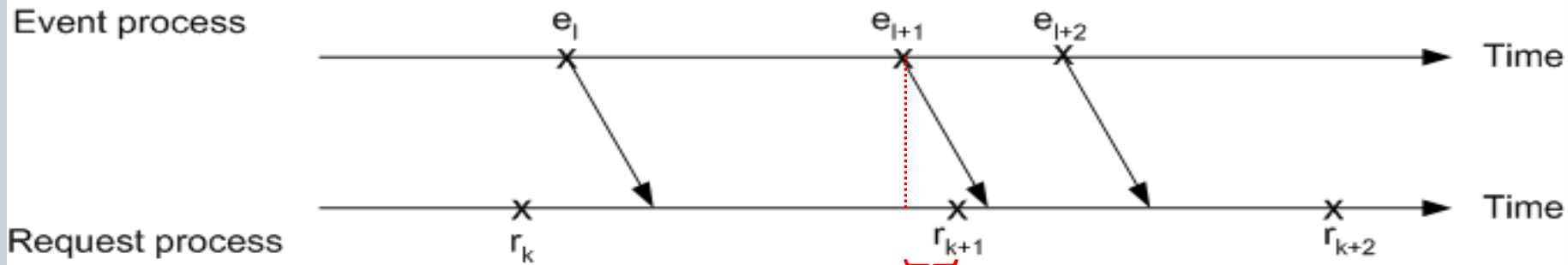
$A(t)$: c.d.f of forward recurrence time

Poisson distributed Event process

$$\begin{aligned}
 \text{mmPr} &= 1 - \int_{t=0}^{\infty} (1 - A(t)) f_D(t) dt \\
 &= 1 - \int_{t=0}^{\infty} e^{-\lambda t} f_D(t) dt \\
 &= 1 - \mathcal{L}\{f_D\}(\lambda)
 \end{aligned}$$

Analysis: Proactive Event-Driven Case

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Pro-active event-driven: Two cases

a) Incremental updates:

$1 - \text{mmPr} = \Pr(\text{no message in transit at } R_k) = \Pr(E/D/\infty \text{ Queue is idle})$

Assuming Poisson Event and delay processes: $\text{mmPr} = \Pr(M/M/\infty \text{ queue is busy}) = 1 - \exp(-\lambda/\nu)$

b) Full updates

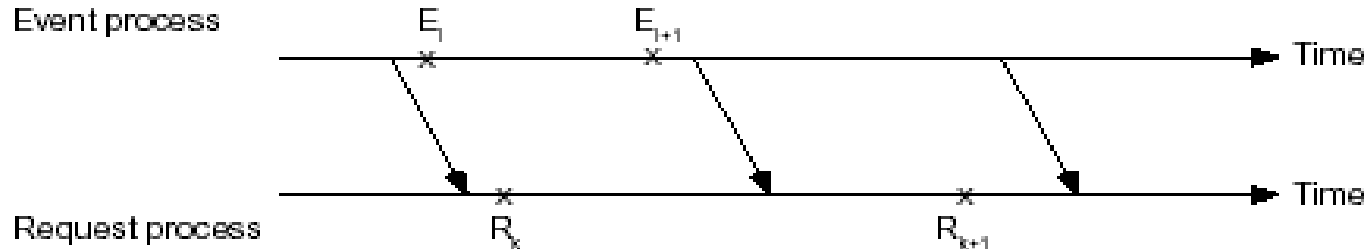
$\text{mmPr} = \Pr(\text{last message still in transit}) = \Pr(E/D/I/I \text{ Queue is busy})$

equivalently:

$$\begin{aligned} \text{mmPr}_{\text{proact,full}} &= 1 - \int_0^\infty P(\mathcal{D} \leq t | U = t) a(t) dt \\ &= 1 - \int_0^\infty P(\mathcal{D} \leq t) a(t) dt. \end{aligned}$$

= same as reactive!!

Analysis: Proactive periodic case



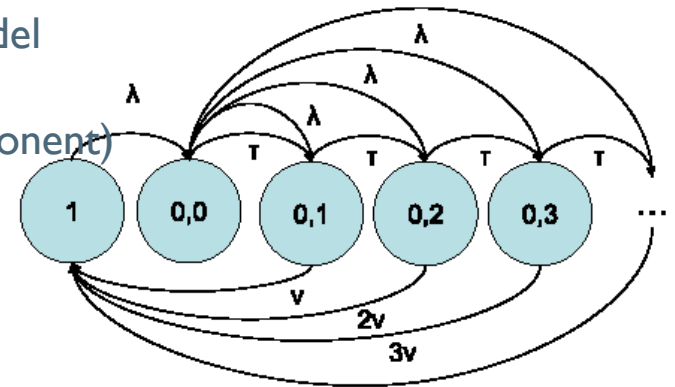
Note: Only the leftmost lambda transition is directed to the right, all the other lambda arrows go to the left and end in state (0,0); all tau arrows transit to the right in the figure

Pro-active periodic:

- Assuming all processes are Poisson
- Representation as continuous-time Markov chain model
- Matching state marked with 1
- Counting #useful messages in transit (in second component)
- Explicit solution:

$$\pi_1 = 1 - \text{mmPr} = \frac{\frac{\nu}{\lambda} \sum_{n=3}^{\infty} (n-2) \prod_{i=1}^{n-2} \frac{\tau}{i\nu + \tau + \lambda}}{1 + \frac{\nu}{\lambda} \sum_{n=3}^{\infty} (n-1) \prod_{i=1}^{n-2} \frac{\tau}{i\nu + \tau + \lambda}}$$

- Alternative representation
(from stochastic argument similar to other cases)



$$\text{mmPr} = \frac{\lambda}{\nu} \exp\left(\frac{\tau}{\nu}\right) \frac{\Gamma\left(\frac{\tau+\lambda}{\nu}\right)}{\left(\frac{\tau}{\nu}\right)^{(\tau+\lambda)/\nu}} F_{\Gamma}\left(\frac{\tau+\lambda}{\nu}, \frac{\tau}{\nu}\right)(1),$$

Quantitative results: Poisson Event Process

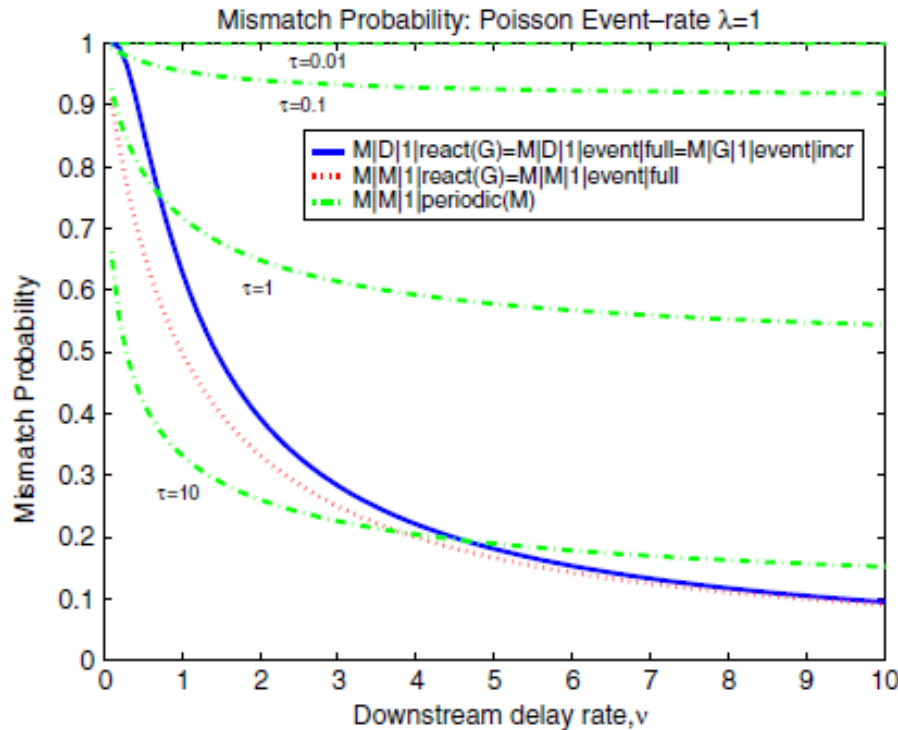


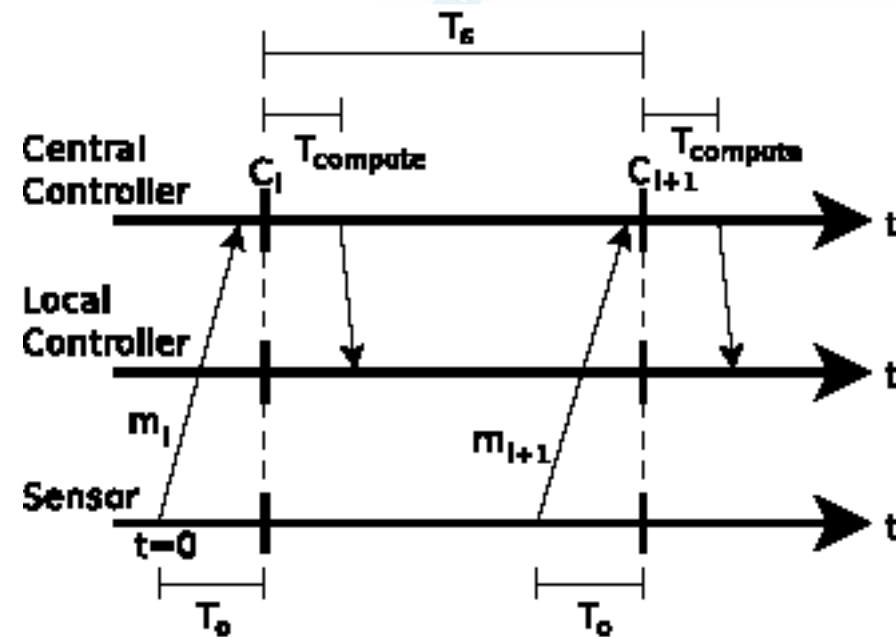
Figure:

- Event process always exponential (first ,M')
- Network delay exponential in dotted and dashed-dotted curves
 - deterministic or more general in solid curve
- Scheme see legend
 - Periodic(M) means exponentially distributed period of update sending

- **Deterministic delay:** reactive and event-driven strategies same mmPR= $1-e^{-\lambda D}$ [blue curve]
- event-drive incremental strategy independent of delay distribution [blue curve]
- **exponential delays** [red curve] better than deterministic for reactive=event-driven full: $\lambda/(\lambda+v)$
- for large v (**fast networks**), periodic strategy [green curves]: mmPr $\rightarrow \lambda/(\lambda+\tau)$, hence worse than others

Wind farm communication deterministic periodic with controlled offset

- Simplified message sequence diagram
 - T_s : Control period (150 ms)
 - $T_{compute}$: computation time (50 ms)
 - T_o : offset
 - C_i : Computation instant
- Compare sensor information at time T_o to sensor information at time C_i
- Higher T_o gives better chance of reaching C_i in time, however larger risk of data changing significantly (larger 'age' of information)



Lecture outline

- Different access scheme
- Performance metrics and results
- **Data quality model for wind-farm scenario**
- Wind-farm scenario using OTS (off-the-shelf) communication technologies—not part of exam
- Summary

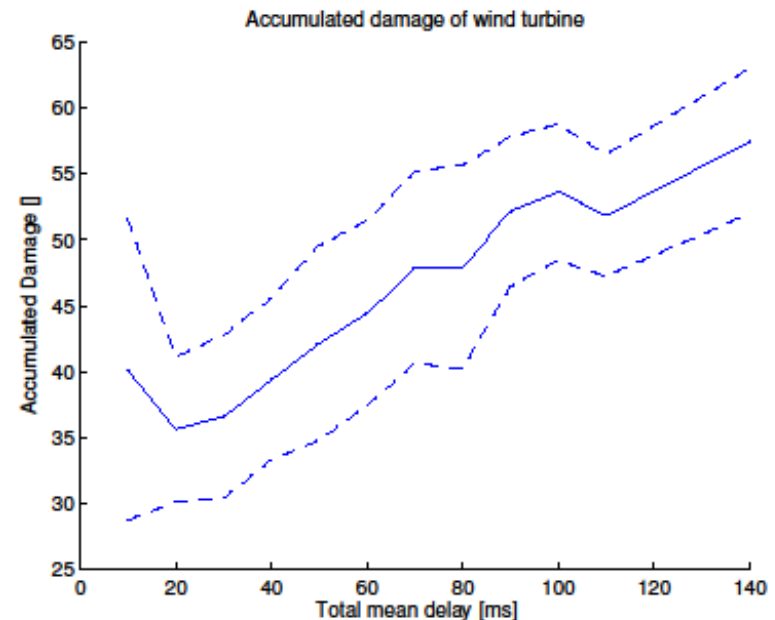
Controller performance results: simulation result for delay

Scenario:

- Increasing mean delay sensor $\rightarrow$ controller (shifted exponentially distributed)
- offset fixed at 75 ms

Observations

- When delays > 20 ms: damage increases with increasing delays



Controller performance results: offset

Scenario:

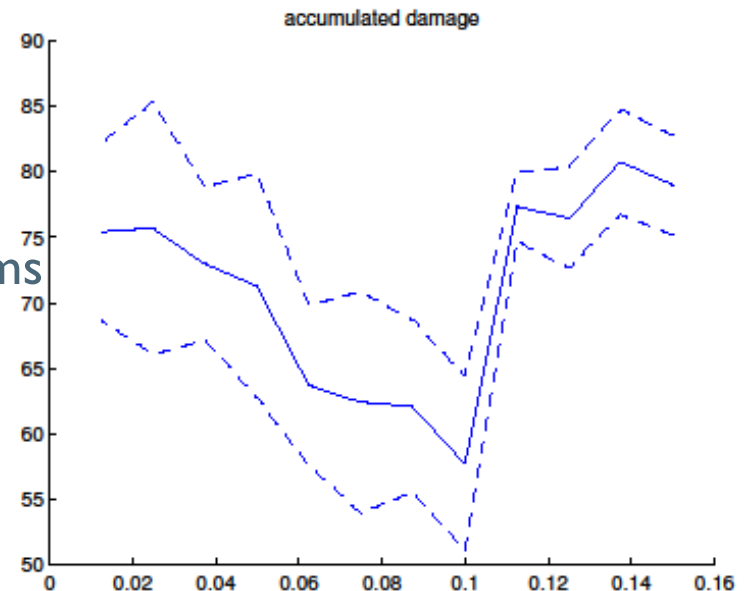
- 30 ms mean exponentially distributed delays, shifted 1 ms
- Vary offset between 10ms and 150ms

Observations:

- Accumulated damage decrease as T_o increases, until $T_o = 100$ ms

Explanation: At $T_o = 100$ ms wind turbine receives previous controller set-point

- This means that at higher T_o 's the old set-point is used for the sensor information



Controller performance: offset optimization

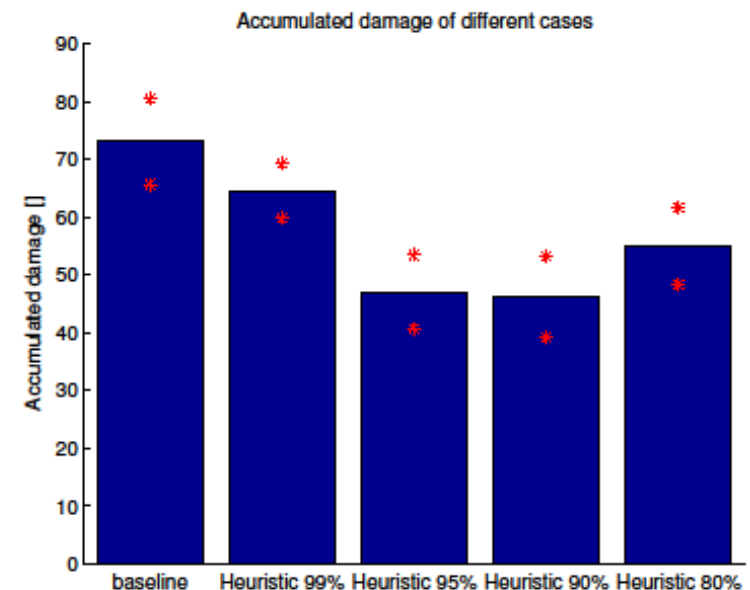
Approaches

- Baseline: uniformly chosen offset in $[0, \dots, 150\text{ms}]$
- Heuristic: take p -percentile of delay distribution
sensor $\rightarrow$ controller

Observation

- Heuristics are efficient, given 'good' choice of p

How to choose p ?



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Break

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Data Quality model motivation

Running simulations to determine, e.g., optimal offset is very time consuming

Basic idea:

- Optimize at a less complex metric, ‘information quality’ reflecting accuracy of information used by controller during control computations:

$$mmPr(\epsilon) := Pr(|X_s(C_i) - X_C(C_i)| > \epsilon)$$

$X_s(C_i)$ ‘true’ value of sensor at time C_i

$X_C(C_i)$ Controller’s knowledge of sensor value at time C_i

Q: Is optimization of mmPr equivalent to optimization of control performance?

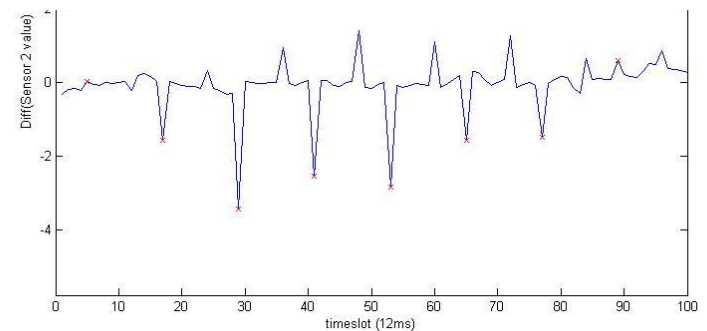
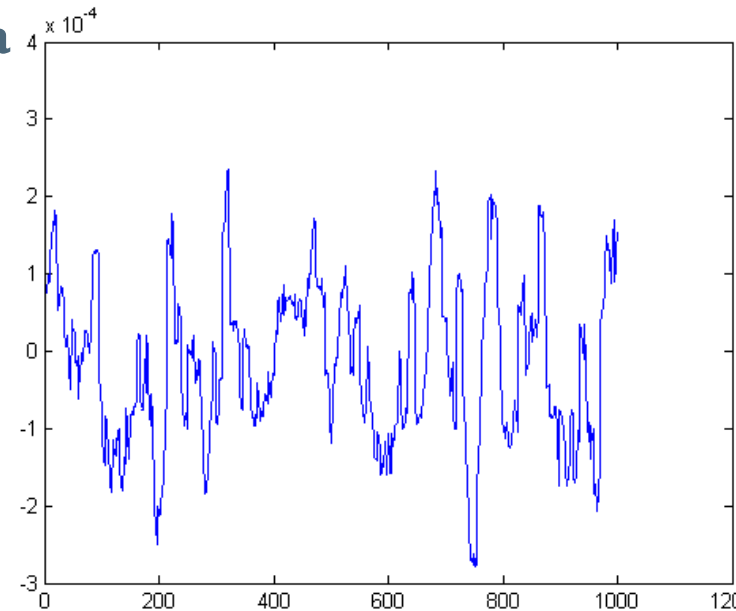
Q: Can we describe mmPr by a simple model?

Modeling the behavior of sensor data

- Run a control simulation for a specific communication scenario (e.g. ideal)
- Take all sensor measurements from this scenario, see snapshot

More detailed inspection:

- sudden changes at control actuations (downward peaks in difference times series)



Markov model

- Approach I: Use discrete time Markov chain
 - Discretize sensor value space ($\rightarrow$ state space size N)
 - Empiric transition probabilities (Matrix P_0) from discretized value trace
- Approach II:
 - Discretize sensor value space
 - Separate data for changes at control actuation: Transition probabilities C
 - Remaining time series: Transition probabilities P

Calculating mmPr

- **Approximation:**

1-mmPr is the probability that the Markov chain is in state i at time C_i , given it was in state i at time $C_i - \text{offset}$ and the update message has been delivered until C_i

- A bit more complex:
consider also the case
that the last update
message does not arrive
in time, see right
(P6 of paper)

$$1 - mmPr = F_D(T_o) \sum_{k=1}^N \pi_u(k) \cdot Pr(\text{Markov chain}$$

is in state k at time instant C_i | Markov chain
has been in state k at time instant $C_i - T_o$)

$$+ (1 - F_D(t_o)) F_D(T_o + T_c) \sum_{k=1}^N \pi_u(k) \cdot$$

$Pr(\text{Markov chain is in state } k \text{ at time instant } C_i |$

Markov chain has been in state k at time instant $C_i - T_o - T_c$)

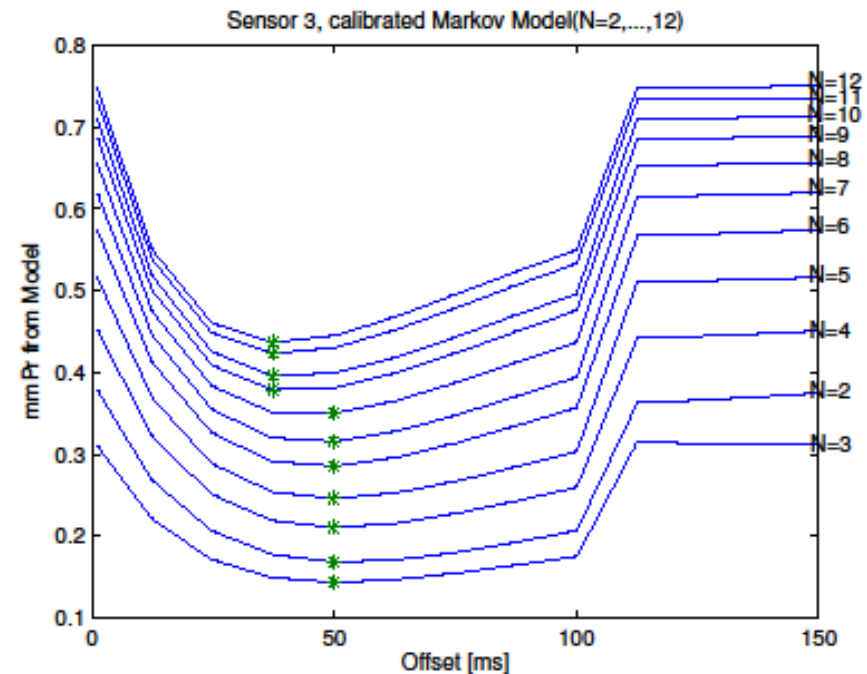
Results

Scenario

- mmPr based on Markov model of sensor 3 data (Approach II)
- Number of states N (discretization bins of sensor info) vary between 2...12

Observations

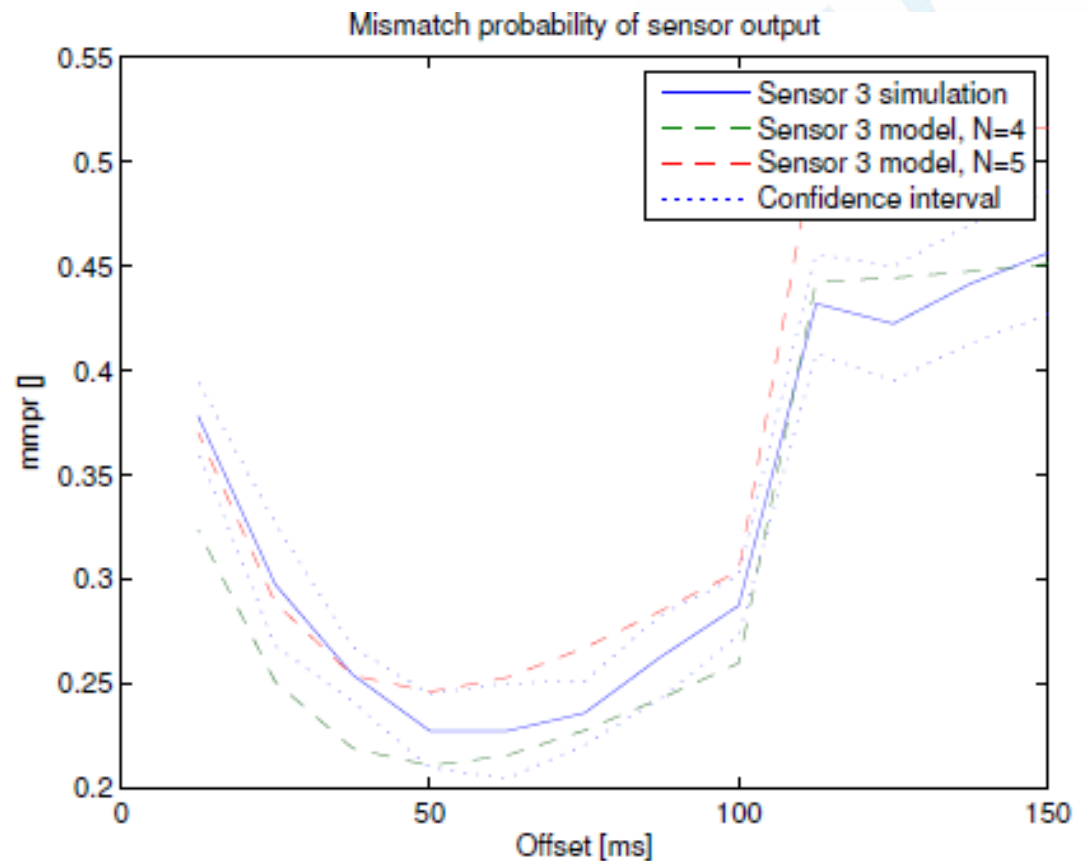
- Curves show minima of mmPr
- ... located at offset of 50ms ($N \leq 8$) respectively 37.5ms ($8 < N \leq 12$)



Validation of model: using mismatch metric

Comparison of simulated mmPr and results from Markov model calculation

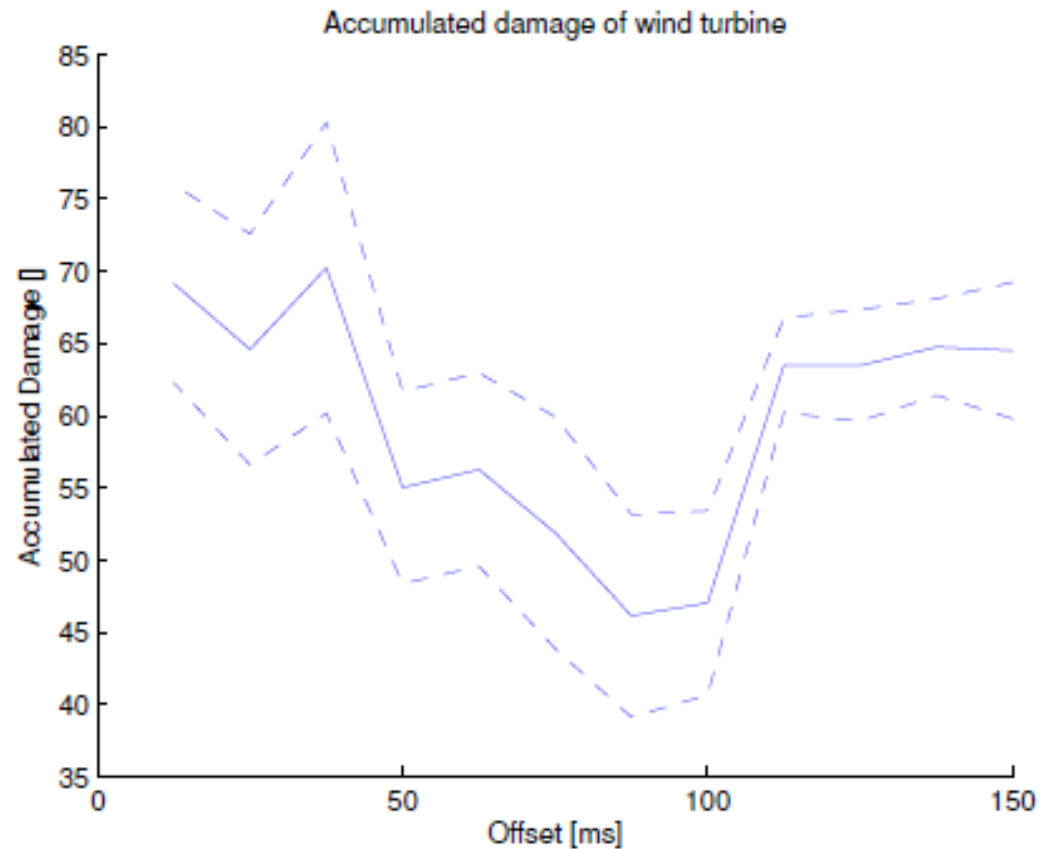
- Qualitative good match
- Minimum occurs at same offset value (50ms in simulation) for N=4 states of Markov chain model



Validation of model: using control performance

Inspection of simulated control performance

→ Discuss graph on the right

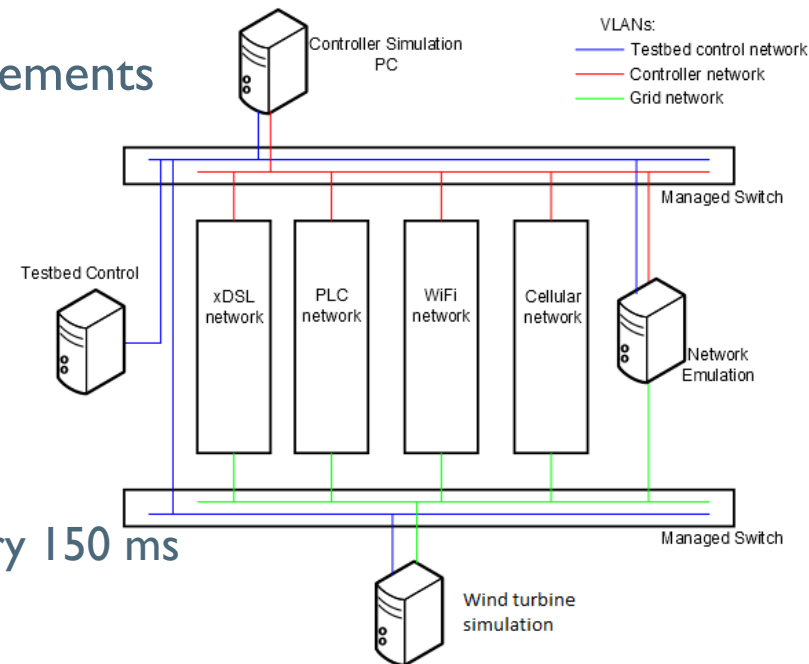


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OTS communication technology measurements

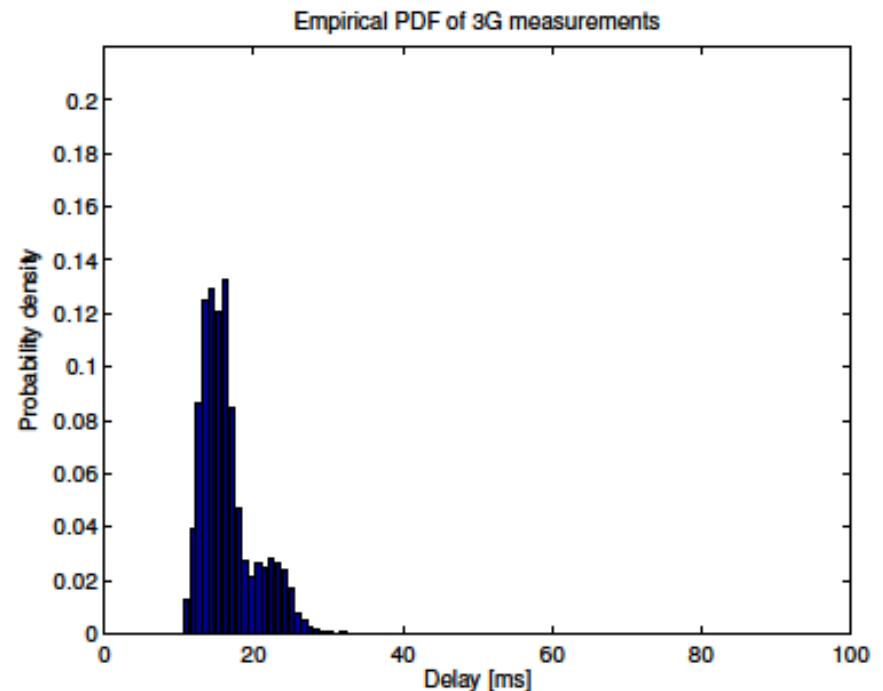
- RTT delay and packet loss measurements for different OTS communication technologies
 - 2G, 3G, PLC and WLAN
- Measurement scenario:
 - 3 sensors at each wind turbine
 - 30 wind turbines in the wind farm
 - Each sensor sends 1 message every 150 ms



- Use the measured delay and packet loss traces for the simulation

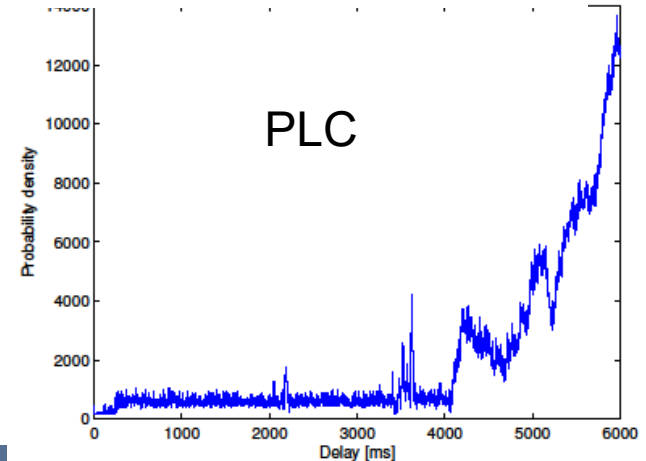
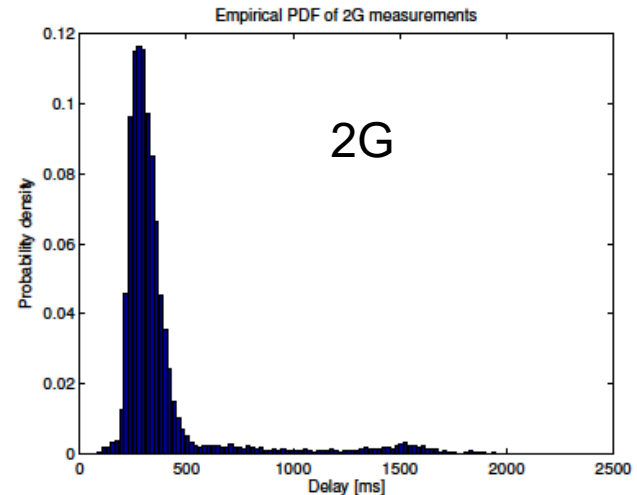
Delay results of OTS communication technologies

- 3G delay results pdf
- RTT delays
- Spike at roughly 15 ms, lower spike at around 25 ms
 - Min delay: 10.5 ms
 - Mean delay: 16.7 ms
 - Max delay: 98.5 ms



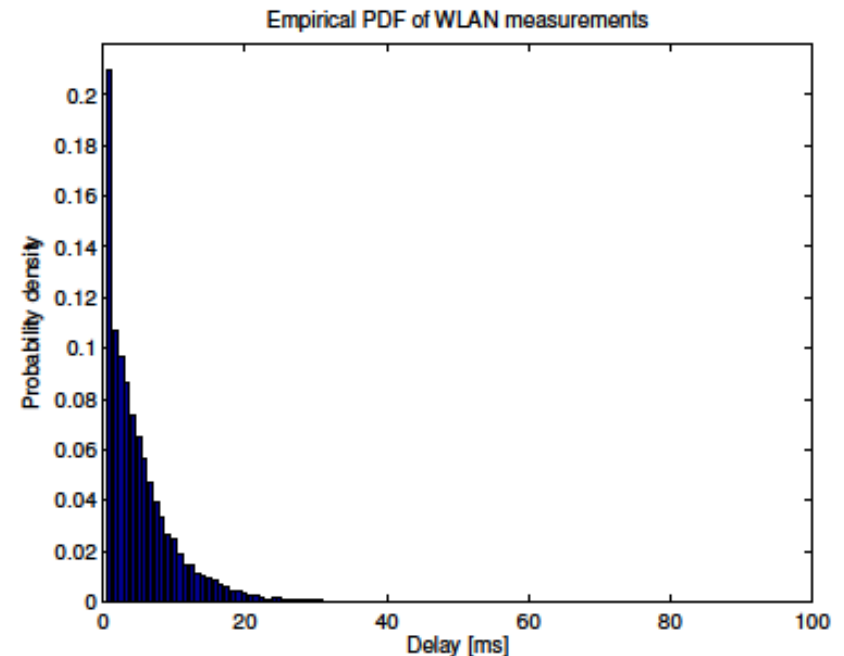
Delay results of OTS communication technologies

- pdf of 2G RTT delays
- Spike at roughly 350 ms
 - Min delay: 84.5 ms
 - Mean delay: 385.2 ms
 - Max delay: 2131 ms
- PLC was not used as it had delays over 3 seconds, and packet losses over 70 %



Delay results of OTS communication technologies

- WLAN shows very low delays, nearly exponentially distributed
 - Min delay: 0.5 ms
 - Mean delay: 5.4 ms
 - Max delay: 83.2 ms



Including measurement results in simulation

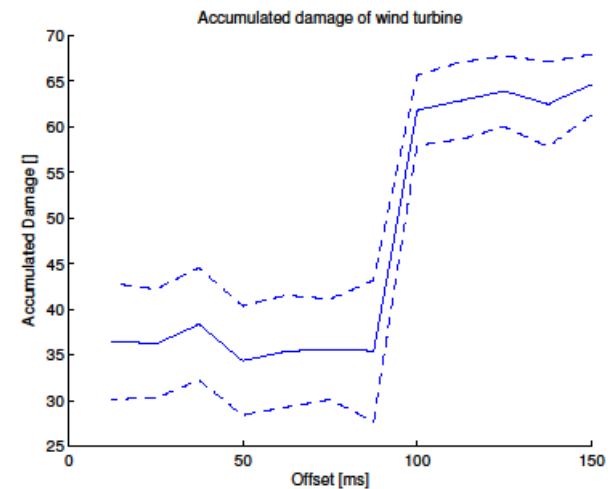
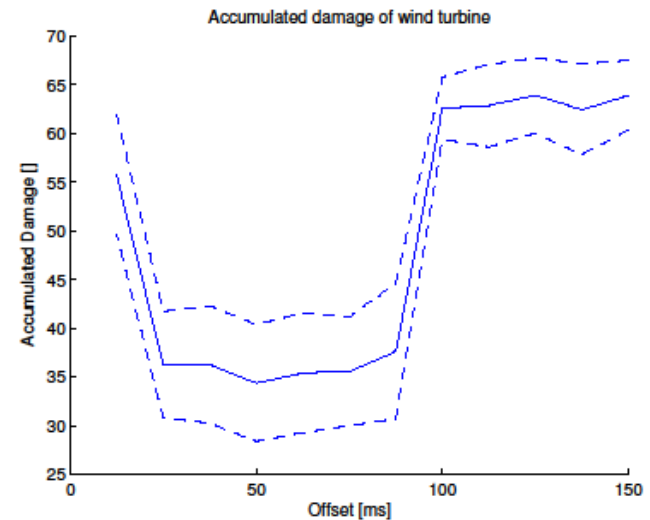
- 2 options for including measurement data in simulation
 - Create distribution based on measurement results, use fitted distribution to determine delays
 - Parametric fit (e.g. Using exponential distributions)
 - Empiric cdf used for sample generation
 - Load all data into network simulator, use the traces directly

Discussion:

- The second option will ensure that any correlation between packet delays are kept but then simulation runs are limited by the size of the measurement!

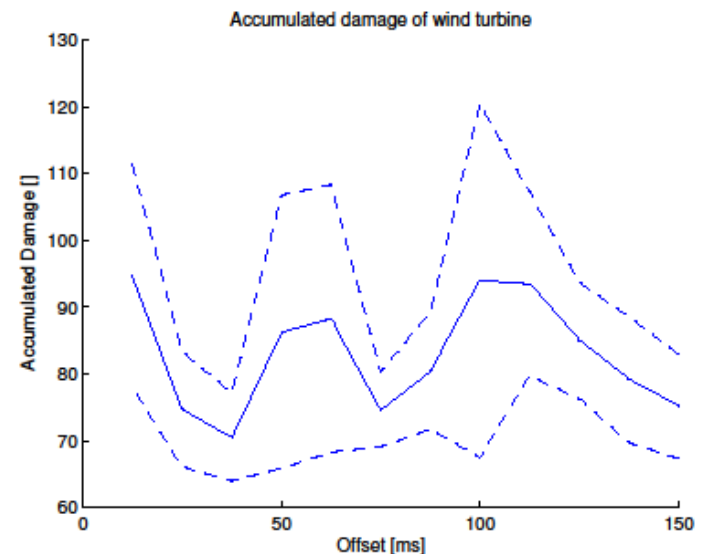
Results

- Top: accumulated damage for 3G trace
 - At low T_o , messages are not received in time
- Bottom: accumulated damage for WLAN trace
 - Even at low T_o messages are received in time



Results

- Accumulated damage using 2G trace
- As delays are too long, messages are always at least one control period old
- Cannot determine optimal offset T_o



Lecture goals

- Understand different access schemes: reactive (pull), proactive (push) periodic or event-driven
- Understand dependencies between control and communication
- Understand and work with metrics characterizing control performance and data quality
- Understand the difficulty of simulation-based parameter optimization, and benefits of analytic models
- Enable the students to apply simple markov models to optimize control systems

Lecture outline

- Different access schemes
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- Break
- Data quality model for wind-farm scenario
- Wind-farm scenario using OTS communication technologies
- **Summary**

References

Bøgsted, M, Olsen, RL & Schwefel, H-P 2010, 'Probabilistic models for access strategies to dynamic information elements', *Performance Evaluation*, vol. 67, no. 1, pp. 43-60. <https://doi.org/10.1016/j.peva.2009.08.015>

Madsen, JT, Findrik, M, Madsen, TK & Schwefel, H-P 2016, 'Optimizing data access for wind farm control over hierarchical communication networks', *International Journal of Distributed Sensor Networks*, vol. 12, nr. 5, 5936235. <https://doi.org/10.1155/2016/5936235>

Exercises

On moodle there is a trace of sensor values as resulting from a wind turbine control simulation. Load this trace into any of the tools that you are familiar with (preferably Matlab) and

1. Plot the time series and marginal distribution of the trace
2. Fit a Markov chain model with N states to this trace (N is a variable, try $N=2...10$)
3. Calculate and plot the mismatch probability as obtained from the simple Markov model for varying offsets
4. Include an exponential delay distribution with mean of 30ms. Assume that there is always a mismatch if the message is not received within the offset period. Plot the same curve as in (3) but now including this network delay.
5. Compare and discuss your results in comparison to Slide 23